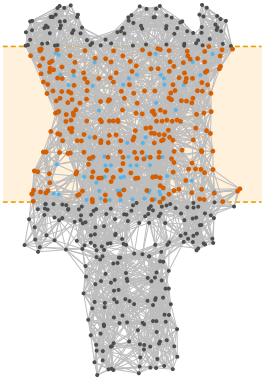
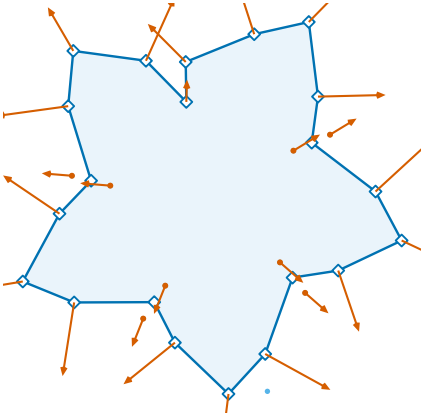
Cα outside the slab membrane Cα loaded outer shell polygon vertex (shell mean) nodal force $\tau\,\partial A/\partial \mathbf{x}$

a MscL Cα truss, r_c = 10 Å



b one 4 Å layer, extracellular view



c

Geometrically nonlinear truss
$U = \frac{k}{2}\sum_e (\ell_e - \ell_e^0)^2$, exact kinematics

Conservative membrane tension
$\Pi = U - \tau A(\mathbf{x})$, $A = \frac{1}{2}\mathbf{x}^T\mathbf{H}\mathbf{x}$

Gauge from the symmetries of the load
A invariant under 4 rigid motions; tilt carries a moment

Verified, certified path
Newton = VFIFE; inertia of $\mathbf{K}_t - \tau\mathbf{H}$ at every step

Departure from linearization
stiffening a, turning $\sin\theta \approx \chi$, snapping n_{neg}

Test against partner structures
TM Cα RMSD vs. linear response and null models